# Predictive Value of Plasma D-Dimer Levels for Cancer-related Stroke: A 3-Year Retrospective Study

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*Background:* Cerebrovascular disease is the second leading cause of central nervous system pathology in cancer patients. Cancer-associated hypercoagulation plays an important role in cancer-related stroke. The present study aims to test whether plasma D-dimer levels could predict comorbid malignancy in patients with ischemic stroke. *Methods:* Five hundred sixteen stroke patients with measured D-dimer levels and who were consecutively admitted to our stroke center from 2009 to 2012 were included. Cancer status was determined by medical chart, and 59 patients were identified to have active cancer. An additional 48 cancer patients with stroke were identified from the hospital database. Several D-dimer cutoff levels were used to predict cancer-related stroke. *Results:* Stroke patients with active cancer had significantly higher D-dimer levels than those without cancer ($P < .001$). The average D-dimer level in stroke patients without cancer was .66 ± 1.83 mg/L, whereas the levels for active cancer patients from the stroke center and hospital database were 5.70 ± 9.63 mg/L and 10.47 ± 12.31 mg/L, respectively. When using D-dimer of .55 mg/L or more and multiple territory infarctions as criteria, the specificity and positive predictive value (PPV) for cancer-related stroke were 99.7% and 92.9%, respectively. When using D-dimer of 5.5 mg/L or more as the cutoff value, the test had a high specificity and PPV regardless the brain magnetic resonance imaging (MRI) findings. Six stroke patients fitting our criteria were confirmed to have occult malignancy after comprehensive cancer survey. *Conclusions:* Extraordinary high D-dimer levels or combining D-dimer and MRI findings may be used as a screening tool to detect malignancy in stroke patients. **Key Words:** Stroke—malignancy—coagulopathy—D-dimer—brain magnetic resonance imaging.
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## Introduction

Cerebrovascular disease is the second leading cause of central nervous system pathology in cancer patients.[1-3] Up to 15% of the cancer patients were reported to exhibit thromboembolic complications at autopsy and 50% had clinical manifestations of stroke.[1] Retrospective studies showed that 6%-12% of ischemic stroke patients have concomitant malignancies.[4,5] According to a 5-year cohort, approximately 20% of cryptogenic stroke patients had active cancer at the time of stroke, whereas additional 10 patients (.3%) were identified with hidden malignancy after a comprehensive work-up.[6]

Growing evidence suggests that cancer-associated hypercoagulopathy might be an important but underestimated risk factor for cancer-related stroke.[6-8] Plasma D-dimer measurements have been used in the diagnosis

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of a variety of thrombosis-related diseases.[9] Plasma D-dimer levels were significantly correlated with activity and prognosis of many cancers.[10-12] Stroke patients with concomitant malignancy were also reported to exhibit significantly higher plasma D-dimer levels than those without cancers.[6,13-15] Furthermore, abnormal D-dimer levels and infarctions over multiple vascular territories were independent predictors of cancer-related stroke.[6,16] However, the D-dimer cutoff level and its predictive value for identifying stroke caused by malignancy remain controversial. The aim of the present study was to test the feasibility of D-dimer levels as a predictive marker for concomitant malignancy in stroke patients.

## Methods

### *Patient Population*

Between January 2009 and January 2012, we retrospectively studied 1361 patients who were admitted to the neurology department of Taichung Veterans General Hospital with the diagnosis of acute ischemic stroke. Using *International Classification of Diseases, Ninth Revision*, diagnostic codes 434.0-434.8 (cerebrovascular diseases) and 140.0-208.0 (malignancy), we identified a further 48 cancer patients with stroke from other departments of our hospital during the same period. Only subjects who received a blood test for plasma D-dimer within 1 month of the stroke event were included (Fig 1).

Three cancer statuses were assigned to each participant according to their medical records. Active cancer was defined as (1) cancer diagnosed within 1 year of stroke occurrence, (2) cancer diagnosed before stroke with incomplete treatment, or (3) recurrence of the prior inactive cancer within 1 year of stroke event. Inactive cancer was defined as a history of cancer before the occurrence of stroke with complete remission. Subjects without any history of cancer before or after the stroke event were defined as "no cancer." We excluded subjects who were lost to follow-up within 1 year of acute stroke because of uncertainty in their cancer status. The investigation conformed to the principles outlined in the Declaration of Helsinki. All study protocols were approved by the local Institutional Review Boards (VGHTC-IRB-CE12132).

Five hundred sixty-four subjects fulfilled our selection criteria and were further divided into 5 subgroups: (1) stroke patients without cancer from SRS, (2) stroke patients with inactive cancer from SRS, (3) stroke patients with active cancer from SRS, (4) stroke patients with inactive cancer from hospital database, or (5) stroke patients with active cancer from hospital database (Fig 1).

### *Laboratory Tests and Brain Images*

Demographic data including age, sex and histories of hypertension, diabetes, hyperlipidemia, atrial fibrillation, and smoking habits were obtained from medical records. Stroke subtypes were assigned using the criteria of the Trial of Org 10172 in Acute Stroke Treatment. Laboratory data including platelet count, prothrombin time, activated partial thromboplastin time, serum levels of fasting blood glucose, total cholesterol, high-density lipoprotein cholesterol, triglyceride, and plasma levels of D-dimer and fibrinogen were collected. All patients received computed tomography of the brain, electrocardiography, and carotid ultrasonography. In addition to brain computed tomography, 237 of the 564 subjects had magnetic resonance imaging (MRI) of the brain. For subjects with brain MRI, the distributions of ischemic lesions were divided into single or multiple vascular territories.

### *Statistical Analyses*

Statistical analyses were performed by SPSS version 19.0. A $P$ value less than .05 was considered statistically significant. All data are presented as the mean ± standard deviation or number (%). Categorical variables were compared by chi-square test, and continuous variables were compared by 1-way analysis of variance. Post hoc analysis (Scheffe test) was used to compare the variables between any 2 of

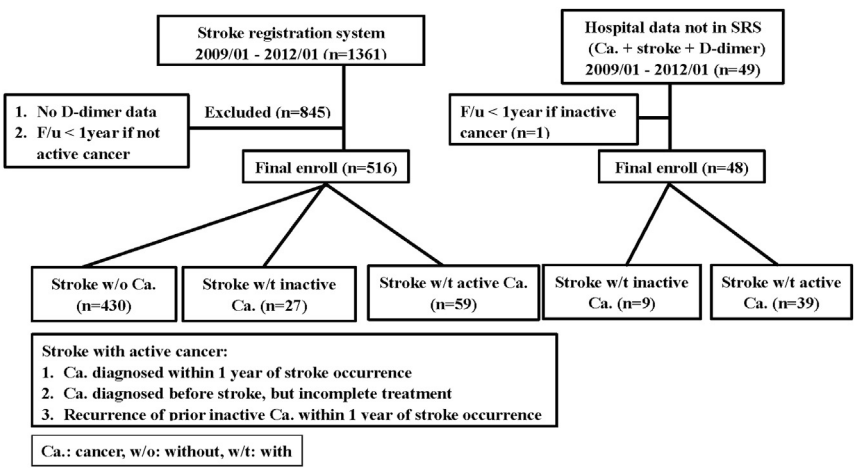


**Figure 1.** *Flow chart of patient selection.*

the 5 subgroups. The optimal D-dimer value for predicting cancer in stroke patients was selected by the receiver operating characteristic scatter plot. Several D-dimer cutoff levels were selected, and their corresponding sensitivity, specificity, positive predictive value (PPV), and negative predictive value were calculated.

## Results

A total of 564 subjects were included in the present study, including 516 from the stroke registration system of our neurology department and 48 from the hospital database (Fig 1). Demographic data and clinical characteristics of each subgroup are shown in Table 1.

Compared with the stroke patients without cancer, subjects with active cancer had significantly lower frequencies of cardiovascular risk factors (Table 1). The plasma levels of D-dimer were significantly higher in the stroke patients with active cancer than those without active cancer ($P < .0001$). The average D-dimer value in stroke patients without cancer was .66 ± 1.83 mg/L, whereas the values for active cancer patients from SRS and hospital database were 5.70 ± 9.63 mg/L and 10.47 ± 12.31 mg/L, respectively (Table 1, Fig 2, A). There was no significant difference in D-dimer levels between patients without cancer and those with inactive cancer ($P = .84$). A higher proportion of active cancer patients had infarctions over multiple territories when compared with noncancer patients ($P < .01$). The active cancer patients exhibited a significantly higher 1-year mortality rate than those without cancer ($P < .01$).

Among the 98 stroke patients with active cancer, 83 had proved by pathology and 15 were diagnosed by image findings. Twenty (20.4%) patients were diagnosed with malignancy after the occurrence of stroke. The most common malignancies related to stroke were gastrointestinal (38.8%), lung (23.5%), and hematological cancers (9.2%). Up to 55.1% of the cancer pathology was adenocarcinoma. Fifty-one patients (52%) had distant metastasis at the time of stroke occurrence. Thrombotic events of other systems were noted in 18 patients. Majority of these patients (61.2%) underwent antiplatelet therapy and only 25.5% of patients received anticoagulants as stroke prevention. Sixty-nine active cancer patients were followed up at our hospital, 7 of whom (10.1%) had recurrent stroke within 1 year of initial event. Forty-nine (71.0%) patients died within 1 year of stroke, including 9 died of massive infarctions over major organs. Except for 1

**Table 1.** *Demographical characteristics of study participants*

| | SRS | | | Hospital database | |
|---|---|---|---|---|---|
| | No cancer (N = 430) | Inactive cancer (N = 27) | Active cancer (N = 59) | Inactive cancer (N = 9) | Active cancer (N = 39) |
| Male (%) | 69.1 | 55.6 | 59.3 | 66.7 | 56.4 |
| Age (y) | 69.0 ± 12.6 | 70.2 ± 13.7 | 72.9 ± 12.6 | 68.7 ± 16.4 | 68.0 ± 14.7 |
| D-dimer (mg/L) | .66 ± 1.83 | .59 ± .60 | 5.70 ± 9.63** | .60 ± .43 | 10.47 ± 12.31** |
| Fibrinogen (mg/dL) | 328.2 ± 85.3 | 301.7 ± 64.6 | 322.8 ± 104.0 | 376.7 ± 100.7 | 296.3 ± 137.6 |
| Platelet ($\times 10^3$/mm$^3$) | 217.0 ± 66.2 | 212.5 ± 77.3 | 190.4 ± 98.2 | 227.2 ± 66.8 | 200.6 ± 147.5 |
| PT (s) | 10.8 ± 2.1 | 11.0 ± 2.7 | 11.9 ± 3.7** | 10.4 ± .3 | 13.4 ± 5.3** |
| aPTT (s) | 28.3 ± 5.1 | 29.1 ± 9.0 | 28.1 ± 5.8 | 27.3 ± 2.4 | 28.9 ± 4.8 |
| Albumin (g/dL) | 3.91 ± 1.60 | 3.85 ± .50 | 3.48 ± .58* | 3.84 ± .27 | 3.29 ± .70* |
| BMI (kg/m$^2$) | 24.78 ± 3.67 | 24.34 ± 2.74 | 23.77 ± 4.37 | 24.43 ± 4.55 | 22.67 ± 3.15** |
| Hypertension (%) | 72.1 | 74.1 | 61.0 | 77.8 | 51.3** |
| Diabetes (%) | 32.1 | 18.5 | 23.7 | 66.7* | 23.1 |
| Hyperlipidemia (%) | 16.0 | 3.7 | 5.1* | 33.3 | 25.6 |
| Smokers (%) | 44.0 | 18.5** | 33.9 | 44.4 | 30.8 |
| Atrial fibrillation (%) | 19.5 | 7.4 | 16.9 | .0 | 7.7 |
| TOAST | | | | | |
| Large artery (%) | 27.2 | 25.9 | 23.7 | 55.6 | 17.9 |
| Cardioembolic (%) | 15.8 | 3.7 | 16.9 | .0 | .0 |
| Small vessel (%) | 37.0 | 48.1 | 30.5 | 11.1 | 5.1 |
| Other determined | 2.6 | 3.7 | 1.7 | 11.1 | .0 |
| Undetermined (%) | 17.4 | 18.5 | 27.1 | 22.2 | 76.9** |
| Multiple territory infarction (n/N$^1$) | 3.3% (6/180) | .0% (0/13) | 28.0% (7/25)** | 16.7% (1/6) | 46.2% (6/13)** |
| Mortality within 1 y (n/N$^2$) | .0% (0/430) | .0% (0/27) | 60.0% (24/40)** | .0% (0/9) | 86.2% (25/29)** |

Abbreviations: aPTT, activated partial thromboplastin time; BMI, body mass index; n/N$^1$, number of patients with multiple territory infarctions/number of patients with available MRI; n/N$^2$, number of patients died within 1 year/number of patients follow-up for more than 1 year or deceased within 1 year; PT, prothrombin time; SRS, stroke registration system; TOAST, Trial of Org 10172 in Acute Stroke Treatment.

*$P < .05$, **$P < .01$, in comparison with the group of "stroke patients without cancer " using analysis of variance or chi-square test.

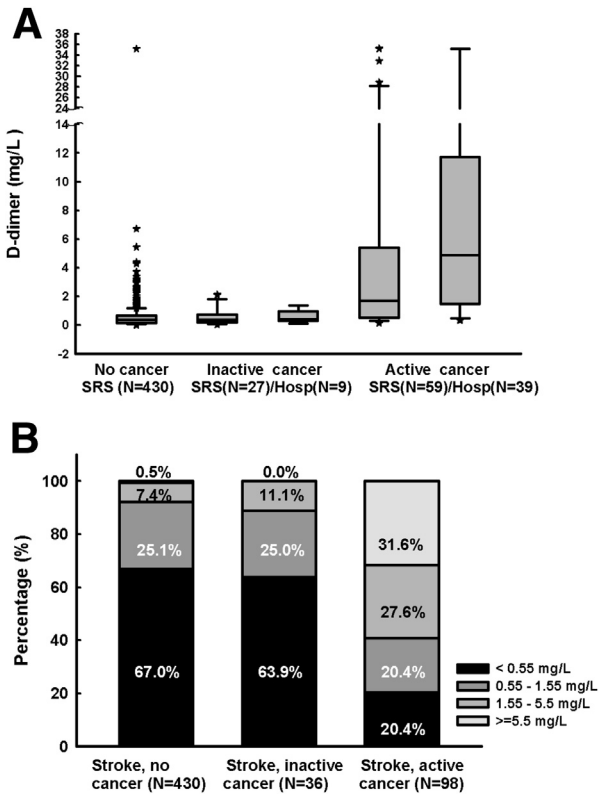


**Figure 2.** *(A) Average D-dimer levels in stroke patients with different cancer status. (B) Distribution of stroke patients using different D-dimer cutoff level.*

person, all cancer-related stroke patients died from cancer-related complications. The mortality rate was 69.8% in the antiplatelet drugs group and 61.1% in the anticoagulants group ($P = .51$).

The normal range of D-dimer level in our hospital was less than .55 mg/L. When D-dimer of .55 mg/L or more was used as the cutoff level, the test had fair sensitivity (79.6%) but inadequate specificity (66.7%, Table 2). We then selected 1.55 and 5.5 as our cutoff levels based on a specificity of 90% and 99%, respectively. When D-dimer of 1.55 mg/L or more was used as the cutoff level, the specificity increased to 91.8% with a modest sensitivity (59.2%). Approximately two thirds of the active cancer patients had D-dimer levels higher than 1.55 mg/L, whereas only 7.9% of the noncancer group had such a high D-dimer value (Fig 2, B). When we included both D-dimer of .55 mg/L or more and multiple territory infarctions as our diagnostic criteria, the specificity increased to 99.7%. However, the sensitivity dropped to 24.5%. Among the stroke patients without cancer, only 1 subject met this criterion. Combining D-dimer of 1.55 mg/L or more and multiple territory infarctions as criteria led to a specificity of 100.0% but with a very low sensitivity (16.7%). When D-dimer of 5.5 mg/L or more was used as the cutoff value, the test yielded a high specificity (99.6%) regardless of brain MRI findings. Only 2 subjects without cancer had a D-dimer value higher than 5.5 mg/L. One had thrombus over the left atrium and the other had septic shock at admission. Comprehensive cancer survey was conducted for 6 stroke patients fitting above criteria. Five patients were found to have hidden malignancy, and 4 of them had distant metastasis at the time of diagnosis. The only 1 with initial negative survey was diagnosed as hepatoma within 1 year.

## Discussion

The aim of the present study was to examine the feasibility of using plasma D-dimer levels to predict concomitant malignancy in stroke patients. A D-dimer level higher than the normal range was not sufficient to diagnose cancer-related stroke as one third of cancer-free stroke patients also had abnormal D-dimer levels (Fig 2, B). For patients who had mildly elevated D-dimer (≥.55 mg/L) and multiple infarctions on MRI, the likelihood of occult malignancy was higher than 99%. The specificity and PPV for a D-dimer level higher than 5.5 mg/L were greater than 93%, which suggested that a comprehensive cancer survey is worthwhile in such patients. Adenocarcinoma at lung and gastrointestinal tracts and hematological malignancies should be carefully excluded when cancer-related stroke was suspected.

**Table 2.** *Using different cutoff levels of D-dimer to predict malignancy in stroke patients*

| | Sensitivity | | Specificity | | PPV | | NPV | | |
|---|---|---|---|---|---|---|---|---|---|
| D-dimer cutoff level | SRS (%) | All (%) | SRS (%) | All (%) | SRS (%) | All (%) | SRS (%) | All (%) | Recommendatory exams |
| ≥.55 mg/L | 72.9 | 79.6 | 66.7 | 66.7 | 22.1 | 33.5 | 95.0 | 94.0 | Brain MRI |
| ≥1.55 mg/L | 50.8 | 59.2 | 91.7 | 91.8 | 44.1 | 60.4 | 93.5 | 91.5 | Brain MRI, serum tumor markers |
| ≥5.5 mg/L | 23.7 | 31.6 | 99.6 | 99.6 | 87.5 | 93.9 | 91.0 | 87.4 | Full survey for occult malignancy |
| ≥.55 mg/L + multi-infarction* | 18.9 | 24.5 | 99.7 | 99.7 | 87.5 | 92.9 | 92.4 | 90.4 | Full survey for occult malignancy |
| ≥1.55 mg/L + multi-infarction* | 13.3 | 16.7 | 100.0 | 100.0 | 100.0 | 100.0 | 91.7 | 88.9 | Full survey for occult malignancy |

Abbreviations: MRI, magnetic resonance imaging; NPV, negative predictive value; PPV, positive predictive value; SRS, stroke registration system; All, overall subjects from SRS and hospital database.

*Multiple territory infarctions detected by brain magnetic resonance imaging.

Elevated plasma levels of D-dimer have been found in cancer-related stroke in several studies.[4,6-8,13-17] When 1 study with only 16 cancer patients was excluded,[4] the average reported D-dimer levels for stroke patients with cancer ranged from 3.6 to 11.5 mg/L,[4,6,7,16,17] which are compatible to levels in our patients with active cancer. Kim et al showed that elevated D-dimer levels and multiple territory infarctions were predictive factors for occult malignancy. In their study, all patients with multiple territory infarctions and a D-dimer higher than 2.15 mg/L were diagnosed with occult malignancy after a full survey.[6]

Cancer patients are predisposed to a prothrombotic state, and D-dimer levels are positively correlated to the disease activity of various cancers.[10-12] Stroke itself is also associated with increased D-dimer levels, especially in cardioembolic subtype.[18-21] The D-dimer levels in our cardioembolic, atherosclerotic, and lacunar stroke patients were 1.16 ± 1.73, 1.10 ± 3.75, and .67 ± 1.25 mg/L, respectively, which explains that extraordinary high D-dimer level without definite etiology might be a hint for hidden malignancies despite stroke subtypes.

Previous studies have shown that multiple ischemic lesions over multiple territories are common in stroke patients with active cancer.[6,13-16] Kim et al[6] reported that 76.1% of cancer-related stroke patients had infarctions over multiple vascular territories on DWI. In our active cancer patients, only one third had infarctions over multiple vascular territories, which is similar to a 10-year cohort study where multiple territory infarctions were found in 24% of cancer-related stroke patients.[7] These data suggest that using MRI features alone has an inadequate sensitivity for prediction of hidden malignancy.

Identifying hidden malignancy in stroke patients is important for improving prognosis prediction and treatment strategies. The published mortality rate in cancer-related stroke ranges from 25% to 30%, compared with 14% in stroke patients without cancers.[3,22,23] In our study, the 1-year mortality rate in patients with active cancer was more than 50%. Patients with cancer-related stroke are predisposed to a high risk of subsequent embolic events in other major organs.[5,22] In our study, 18.4% of cancer-related stroke patients developed embolisms over other organs. For stroke patients with cancer, the treatment should be focused on underlying diseases.[2,15,23,24] There was no significant difference about recurrence of embolic events and mortality rate in our patients with different preventive regimens.

There are several limitations in our study. First, this is a retrospective study. We acknowledged that the distributions of cardiovascular risk factors were different between stroke patients with cancer and those without cancers. However, D-dimer levels remained to be the most significant predicting factor of cancer after adjusting for other cardiovascular risk factors (supplemental Table S1). Besides, stroke patients with active cancer malignancy had significantly higher D-dimer levels than those without active cancer regardless of stroke subtypes (supplemental Table S2). Second, we did not include cancer patients without stroke for analysis, and it remains unclear whether elevated D-dimer levels are the result of the stroke or the cancer. Third, not all patients with high D-dimer levels received malignancy surveys in our study, which may affect the accuracy of the D-dimer test in predicting stroke patients with malignancy.

In conclusion, plasma D-dimer levels may be useful as a screening tool for concomitant malignancy in stroke patients. But further prospective studies are warranted to confirm the optimal D-dimer cutoff values.

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## Supplementary Data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jstrokecerebrovasdis.2013.10.022.

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